IEEE
CAIRO UNIVERSITY STUDENT BRANCH
IEEE
CUSB

ieee**cusb**

Embedded AVR Work **Content**

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Session 1 – Embedded Systems Concepts

Session Content

- What is Embedded Systems?
- Microcontroller vs Microprocessor
- AVR family overview
- Memory map (Flash / SRAM / EEPROM)
- Registers
- Datasheet reading
- Clock system
- Polling vs Interrupt

Session 2 – C Programming (Part 1)

Session Content

- GitHub Intro
- Data types
- Operators
- if / switch
- for / while loops
- Bitwise operations

Session 3 – C Programming (Part 2)

Session Content

- Functions
- Arrays
- Pointers
- Structs
- Macros (#define, #if)
- Header files
- Header guards

Session 4 – GPIO + Layered Architecture

Session Content

- MCAL vs HAL vs APP
- GPIO registers
- Input vs Output
- Pull-up vs Pull-down

Session 5 – LCD & Keypad

Session Content

- LCD interfacing
- Keypad scanning

Session 6 – Interrupts

Session Content

- ISR concept
- External interrupts
- Global interrupt enable

Session 7 – Timers & ICU

Session Content

- Timer modes
- Prescalers
- Input capture

Session 8 – PWM & ADC

Session Content

- Analog vs Digital
- PWM
- ADC

Session 9 – UART

Session Content

- Serial communication
- Baud rate
- TX / RX

Session 10 – I2C & SPI

Session Content

- I2C vs SPI
- Master / Slave
- Real devices